
Where Criticism Ends and Hate Begins: Counterfactual Calibration Invariance for Antisemitism Detection in Social Media Text

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Abstract

We study antisemitism detection on the GOLDSTANDARD2024 dataset under three evaluation axes: positive-class performance, calibration, and counterfactual robustness under identity-term swaps. We first compare prompted instruction-tuned LLMs with fine-tuned transformer classifiers and find that prompting alone is not sufficient for this task. We then introduce Counterfactual Calibration Invariance (CCI v2), a training-time objective that combines focal loss with a Jensen-Shannon consistency penalty between a tweet and its identity-swapped counterfactuals. Across several transformer backbones, CCI v2 substantially reduces hard counterfactual flip rates while preserving, and in some cases improving, positive-class F1. We also show that post-hoc calibration is not sufficient, as Temperature Scaling improves calibration without reducing hard flips, while multicalibration reduces counterfactual stability.

1. Introduction

Antisemitism detection is more difficult than generic hate-speech classification because the boundary between hate speech and legitimate criticism is often subtle. Under the IHRA working definition [1], criticism of Israeli policy is not automatically antisemitic, while antisemitic content may also appear through stereotypes, coded language, or conspiracy narratives [2].

This makes the task challenging for two reasons. First, models must rely on context rather than isolated words. Second, datasets collected through identity-related keywords can make these terms look artificially predictive of the label. As a result, models may become overly sensitive to identity words and unstable under counterfactual substitutions.

We study this problem in three steps: prompted LLMs as an

off-the-shelf baseline, fine-tuned transformers as the main supervised approach, and Counterfactual Calibration Invariance (CCI v2) as a training-time method to improve robustness. CCI v2 trains the model to predict the correct label while remaining consistent across identity-swapped counterfactuals.

2. Related Work

Dixon et al. [3] introduced FPED/FNED and documented identity-term shortcuts as a central failure mode of hate-speech corpora. Keyword balancing [4] acts at the sampling level and does not constrain the learned function on individual pairs. Garg et al. [5] penalise the L_p logit difference between a sentence and its identity-swapped counterpart as a form of approximate individual fairness; Davani et al. [6] extend this idea to hate speech. These approaches operate in logit space, do not explicitly model subgroup-conditional rescaling, and have mainly been validated within a single backbone family.

3. Method

Dataset. We use GOLDSTANDARD2024 [7], a dataset of 11,311 English tweets annotated for antisemitic content. The dataset is imbalanced, with about 17.3% positive examples. The test split contains 1,060 tweets, including 187 positives. It is organized around keyword groups such as *Jews*, *Israel*, *Kikes*, and *ZioNazi*, which have very different positive-class base rates.

Setup. Let x be a tweet. A classifier produces logits $(z_0(x), z_1(x))$, we work with the decision logit $z(x) = z_1(x) - z_0(x)$ and posterior $p(x) = \sigma(z(x))$.

Counterfactual construction. For tweets containing explicit Jewish religious or ethnic terms, we generate up to five counterfactual versions by replacing the identity term with a term from a religiously symmetric vocabulary such as *Muslim*, *Christian*, *Hindu*, *Buddhist*, or *Atheist*. A symmetry filter removes cases where the label is unlikely to be preserved, such as slur-based examples or purely geopolitical

mentions without a religious-ethnic anchor.

Metrics. We evaluate models along three axes: positive-class performance (F_1^+), calibration (Expected Calibration Error, ECE), and counterfactual robustness (Hard and Soft Counterfactual Flip Rates, CFR_{hard} and CFR_{soft}). Let x' be a counterfactual rewrite of x obtained by swapping an identity term, the counterfactual flip rates are $\text{CFR}_{\text{hard}} = \mathbb{P}[(p(x) \geq 0.5) \neq (p(x') \geq 0.5)]$ and $\text{CFR}_{\text{soft}}^{L_i} = \mathbb{E}[|p(x) - p(x')|]$.

Prompted LLMs. We first evaluate prompted LLMs because they are the simplest way to approach the task without supervised training. We test Llama-3.1-8B, Mistral-7B, and Qwen-2.5-7B under zero-shot, few-shot, and Guided Chain-of-Thought prompting. This track answers a practical question: how far can we get with prompting alone?

Fine-tuned transformers. We then study fine-tuned transformer classifiers. The main baselines are RoBERTa-base, DeBERTa-v3-base, HateBERT, and Model C, our strongest compact baseline, which combines DeBERTa-v3-base with focal loss and keyword masking.

Focal loss is useful because the dataset is imbalanced and the hardest cases are often the most informative. Keyword masking is used to reduce direct dependence on identity terms by forcing the model to rely more on context.

CCI v2 loss. CCI v2 extends Model C by adding an explicit counterfactual consistency constraint. The loss was designed around three requirements: the model must still solve the original task, it must remain suitable for imbalanced data, and it should give similar predictions to a tweet and its valid counterfactuals.

For a training example x_i with label y_i and counterfactuals $\{x'_{i,1}, \dots, x'_{i,K_i}\}$, we minimize

$$\mathcal{L}_{\text{CCI}}(x_i) = \ell_{\text{focal}}(f(x_i), y_i) + \frac{\lambda}{K_i} \sum_{k=1}^{K_i} \text{JS}_b(p(x_i), p(x'_{i,k})).$$

The first term preserves the main classification objective. The second term penalizes probability differences between a tweet and its counterfactuals. We use Jensen-Shannon divergence because it is symmetric, bounded, and its gradient vanishes on fair pairs while pulling disagreeing pairs toward their Bernoulli mean. CLP-style L_p logit penalties, by contrast, grow unboundedly with confidence and suppress gradient exactly where the constraint is most violated. The coefficient λ controls the balance between task performance and counterfactual invariance.

Why post-hoc Temperature Scaling is not enough. Temperature Scaling divides logits by a positive scalar after training [8]. This can improve calibration, but it does not change the sign of the decision logit.

Because $T > 0$ preserves the sign of $z(x)$, replacing $z(x)$ by $z(x)/T$ leaves all hard predictions at threshold 0.5 unchanged. Temperature Scaling can therefore improve calibration but cannot reduce CFR_{hard} ; any post-hoc remedy for hard flips must be non-monotone or per-cell.

4. Validation

Table 1 summarizes the main test results on GOLDSTANDARD2024.

Prompted LLMs remain below fine-tuned transformers.

Prompted LLMs improve with stronger prompting, but they do not match fine-tuned transformers on this task. The best prompted result is Llama-3.1-8B with Guided Chain-of-Thought at about $0.45 F_1^+$, clearly below the best supervised models.

This result is important because it sets the starting point of the study. Prompting alone provides a simple and flexible baseline, but it does not offer the level of task specialization needed for this benchmark. In particular, the gap suggests that antisemitism detection on GOLDSTANDARD2024 still benefits strongly from supervised fine-tuning on labeled in-domain data.

CCI v2 improves the strongest supervised baseline.

Among fine-tuned transformers, CCI v2 gives the best trade-off between task performance and counterfactual robustness. The most important comparison is between Model C and CCI v2 on DeBERTa-v3-base: CCI v2 improves F_1^+ from 0.660 to 0.677 while reducing CFR_{hard} from 0.0358 to 0.0183, a 49% reduction in hard counterfactual flips.

This pattern also transfers to larger and different backbones. On DeBERTa-v3-large, CCI v2 reduces CFR_{hard} from 0.042 to 0.019, a 55% reduction, while achieving the best overall F_1^+ in the study. On RoBERTa-large, it reduces CFR_{hard} from 0.047 to 0.013, a 73% reduction and the lowest hard flip rate overall. These results show that CCI v2 does not improve robustness by sacrificing task performance. Instead, it preserves, and in some settings improves, positive-class detection while reducing sensitivity to identity swaps across model families.

CCI v2 reduces flips by compressing scores, not by changing token reliance.

Integrated Gradients show that identity-token attribution mass barely moves: the shift is statistically significant but far too small to account for the robustness gain. Instead, the consistency term compresses the near-boundary score distribution, moving the median positive-class probability from 0.40 to 0.13 preserving F_1^+ . The gain is therefore bought by less inflated scores rather than by a change in which tokens the model reads. The same conservatism reappears off-distribution and underlies the threshold-shift effect below.

Method	$F_1^+ \uparrow$	ECE $\downarrow$	CFR $_{\text{hard}} \downarrow$	CFR $_{\text{soft}}^{L_1} \downarrow$	FNED $\downarrow$
C: DeBERTa+Focal+KWMask	.660 $\pm$.019	.077 $\pm$.034	.036 $\pm$.008	.032 $\pm$.007	.797 $\pm$.159
C + TempScaling	.660 $\pm$.019	.045 $\pm$.014	.036 $\pm$.008	.033 $\pm$.009	.797 $\pm$.159
C + Multicalibration	.652 $\pm$.015	.035 $\pm$.007	.033 $\pm$.006	.063 $\pm$.019	.956 $\pm$.081
Davani CLP ($\lambda=0.5$)	.672 $\pm$.015	.094 $\pm$.033	.040 $\pm$.005	.036 $\pm$.005	.786 $\pm$.036
CCI v2 (DeBERTa-base)	.677 $\pm$.008	.098 $\pm$.053	.018 $\pm$.003	.014 $\pm$.002	.793 $\pm$.180
CCI v2 (DeBERTa-large)	.719 $\pm$.012	.099 $\pm$.063	.019 $\pm$.009	.017 $\pm$.003	.647 $\pm$.058
CCI v2 (RoBERTa-large)	.694 $\pm$.005	.077 $\pm$.052	.013 $\pm$.006	.013 $\pm$.002	.759 $\pm$.115
LLM Llama-3.1-8B, Guided-CoT	.450	—	—	—	—
LLM Mistral-7B, best prompt	.417	—	—	—	—
LLM Qwen-2.5-7B, best prompt	.373	—	—	—	—

Table 1. Test metrics on GOLDSTANDARD2024 ($n=1,060$); CFR $_{\text{hard}}$ and CFR $_{\text{soft}}^{L_1}$ over $\mathcal{P}$ ($|\mathcal{P}|=2,117$). Encoders: mean $\pm$ std over six seeds (five on DeBERTa-large); LLMs deterministic ($T=0$, 4-bit). $\uparrow / \downarrow$: higher/lower better; bold = per-column winner.

Calibration and counterfactual robustness. Temperature Scaling improves calibration but leaves CFR $_{\text{hard}}$ unchanged, consistent with the sign-preservation argument above. On Model C, ECE decreases from 0.077 ± 0.034 to 0.045 ± 0.014 , while CFR $_{\text{hard}}$ remains unchanged at 0.0358 ± 0.008 . This confirms an important conceptual point: post-hoc rescaling can make probabilities more reliable without changing binary decisions at the default threshold.

Multicalibration is more nuanced. We use a post-hoc multicalibration correction following Hébert-Johnson et al. [9], which adjusts predicted probabilities to improve subgroup-conditional calibration after training. In our results, it further reduces ECE to 0.035 ± 0.007 and slightly lowers CFR $_{\text{hard}}$ to 0.033 ± 0.006 , but it substantially worsens soft counterfactual instability, increasing CFR $_{\text{soft}}$ from 0.032 ± 0.007 to 0.063 ± 0.019 . In other words, a model can become better calibrated at subgroup level while becoming less stable across counterfactual pairs. This shows that calibration and counterfactual fairness are related but distinct objectives.

Cross-dataset transfer reveals a threshold-shift effect.

We further evaluate transfer on the Jewish-target subsets of HateXplain and ToxiGen, two external hate-speech benchmarks with different label distributions and collection settings. On these datasets, CCI v2 preserves ranking quality: its PR-AUC differs from Model C by at most 0.021. However, at the default threshold of 0.5, it underperforms in 4 of the 6 architecture–dataset settings. More precisely, CCI v2 tends to produce more conservative scores outside the training distribution. Once the decision threshold is re-tuned on the target dataset, CCI v2 wins or ties in 5 of the 6 settings. The model still ranks examples well, but the default decision threshold becomes suboptimal.

5. Conclusion

This study leads to three main conclusions. First, prompted LLMs are useful reference systems, but they remain clearly below fine-tuned transformers on this benchmark. Second, counterfactual robustness does not follow automatically

from better task performance or post-hoc calibration. Third, CCI v2 provides an effective training-time solution: it adds a simple pairwise consistency term to a strong supervised baseline and substantially reduces hard counterfactual flips while preserving, and in some settings improving, overall performance.

These results should nevertheless be interpreted with care. Our experiments are limited to English tweets, one dataset, and a handcrafted counterfactual vocabulary, so the findings may not transfer directly to other platforms, languages, or forms of coded antisemitism.

Furthermore, our labels follow the IHRA-oriented definition used in GOLDSTANDARD2024, which is appropriate for this benchmark but not conceptually neutral. Other frameworks may disagree on the status of some borderline Israel-related statements.

References

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